RevIN Experiment 3: Centering and Transform Variants

Thesis experiment note

1 Theoretical overview and goal

Mean centering assumes the look-back average is the appropriate location anchor for the future. Last-value centering instead predicts changes around the most recent observation, which may better suit local trends. The inverse hyperbolic sine transform

$$z = \operatorname{asinh}(x)$$

is approximately linear near zero and logarithmic for large magnitudes, giving a reversible compression of heavy-tailed values before instance normalization.

Experimental goal. Test two lightweight changes to RevIN's inductive bias: a more local location anchor and a nonlinear magnitude compression, without introducing the cmIN or distribution-distance experiments.

2 Implementation details

The variants use non-affine instance normalization and normalized MSE:

Method	Center	Transform
`instance_nmse`	Look-back mean	Identity
`revin_last_nmse`	Last observation	Identity
`revin_arcsinh_nmse`	Look-back mean	asinh

Every transformation is inverted before data-space evaluation. The same six datasets, seven settings, DLinear/PatchTST models, five seeds, optimizer, batch size, epoch budget, and validation frequency are used. Configuration mapping is defined by `method_args` in `src/slurm/run_revin_experiment.slurm`; layer behavior is in `src/utils/normalizations.py`.

Tables are separate by model and test split. They report ordinary MSE as mean $\pm$ sample standard deviation with explicit $\times 10^m$ row scaling. The baseline for both contrasts is `instance_nmse`.

3 Interpretation

Last centering tests sensitivity to local level evolution; asinh tests sensitivity to tails and multiplicative scale. A gain is evidence for that specific preprocessing bias, not a general solution to conditional shift. Results should be examined across datasets because either transform may remove useful absolute-level information.